

Quiz 5.1 Sequential Logic Circuit 1																																		
Course Code: CPE 013	Program:																																	
Course Title: Logic Circuits and Design	Date Performed: 31/03/26																																	
Section: 22S2	Date Submitted: 31/03/26																																	
Name: Bulaong, Ross Andrew D.	Instructor: Engr. Noel Lusung																																	
Instructions																																		
Design a 3- bit Synchronous Sequential Circuit using JK flip-flop displayed in LEDs. Sequence should be: 0->7->6->5->4->3->2->1>0 Requirements: - State Table - K-map and Minterm solution - Hand-Written Sequential Circuit Drawing																																		
Answer																																		
State Table <table style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 50%; text-align: center;">Present State (Q2 Q1 Q0)</th> <th style="width: 50%; text-align: center;">Next State (Q2 Q1 Q0)</th> </tr> </thead> <tbody> <tr><td>000</td><td>111</td></tr> <tr><td>001</td><td>000</td></tr> <tr><td>010</td><td>001</td></tr> <tr><td>011</td><td>010</td></tr> <tr><td>100</td><td>011</td></tr> <tr><td>101</td><td>100</td></tr> <tr><td>110</td><td>101</td></tr> <tr><td>111</td><td>110</td></tr> </tbody> </table> K-Map J0 & K0 <table style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 10%;">Q2 \ Q1Q0</th> <th style="width: 20%;">00</th> <th style="width: 20%;">01</th> <th style="width: 20%;">11</th> <th style="width: 20%;">10</th> </tr> </thead> <tbody> <tr> <td>0</td> <td style="text-align: center;">1</td> <td style="text-align: center;">1</td> <td style="text-align: center;">1</td> <td style="text-align: center;">1</td> </tr> <tr> <td>1</td> <td style="text-align: center;">1</td> <td style="text-align: center;">1</td> <td style="text-align: center;">1</td> <td style="text-align: center;">1</td> </tr> </tbody> </table> J1		Present State (Q2 Q1 Q0)	Next State (Q2 Q1 Q0)	000	111	001	000	010	001	011	010	100	011	101	100	110	101	111	110	Q2 \ Q1Q0	00	01	11	10	0	1	1	1	1	1	1	1	1	1
Present State (Q2 Q1 Q0)	Next State (Q2 Q1 Q0)																																	
000	111																																	
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110	101																																	
111	110																																	
Q2 \ Q1Q0	00	01	11	10																														
0	1	1	1	1																														
1	1	1	1	1																														

	Q2 \ Q1Q0	00	01	11	10
0	1	0	0	1	
1	1	0	0	1	
K1					
	Q2 \ Q1Q0	00	01	11	10
0	1	0	0	1	
1	1	0	0	1	
J2					
	Q2 \ Q1Q0	00	01	11	10
0	1	0	0	0	
1	1	0	0	0	
K2					
	Q2 \ Q1Q0	00	01	11	10
0	1	0	0	0	
1	1	0	0	0	
Minterm Solution Table					
	Flip-Flop	J Input		K Input	
Q2		Q1'Q0'		Q1'Q0'	
Q1		Q0'		Q0'	
Q0		1		1	

